# CASO SUMA

4 3 2 1 → Tenemos 3 procesos y uno Extra

101 A
110 B
1011 → S

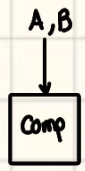
cambia segun

Existen condicionantes

Bueno identificamos las Entradas
↳ A, B, Cin

Salidas
↳ Co, S

| | Entradas | | | Salidas | |
|---|---|---|---|---|---|
| Caso 1 | A=1 | B=1 | Cin=0 | Co=1 | S=0 |
| Caso 2 | A=0 | B=0 | Cin=0 | Co=0 | S=0 |
| Caso 3 | A=1 | B=0 | Cin=0 | Co=1 | S=1 |
| Caso 4 | A=0 | B=1 | Cin=0 | Co=1 | S=1 |
| Caso 1 | A=1 | B=1 | Cin=1 | Co=1 | S=1 |
| Caso 2 | A=0 | B=0 | Cin=1 | Co=1 | S=0 |
| Caso 3 | A=1 | B=0 | Cin=1 | Co=1 | S=0 |
| Caso 4 | A=0 | B=1 | Cin=1 | Co=1 | S=0 |




Entonces tenemos casos que cuando

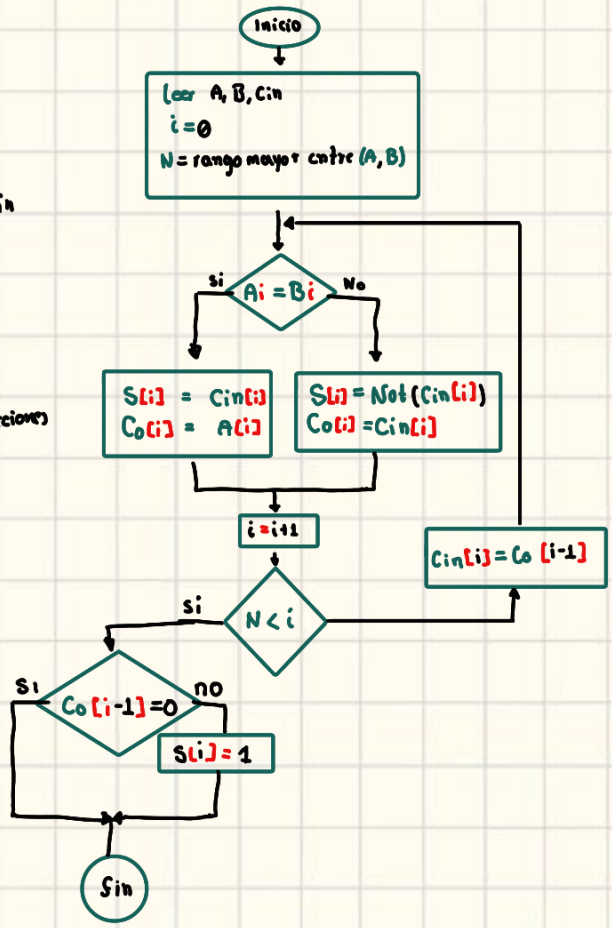
A=B
- A=0: S=Cin, Co=0=A
- A=1: S=Cin, Co=1=A

$\boxed{S=Cin \quad Co=A}$

A≠B
- Cin=0: S=1, Co=0=Cin
- Cin=1: S=0, Co=1=Cin

$\boxed{S=\text{inversa cin} \quad Co=Cin}$

logramos simplificar las desiciones
↳ pasamo de do 8 acciones a tener so 2 acciones




jamolo a prueva Suma → (10110) → A
(10010) → B

| B | Cin | Cout | S |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 1 | 0 | 1 | 0 |
| | | | 1 |

| i | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|
| | 1 | 0 | 1 | 0 | 0 | 0 |

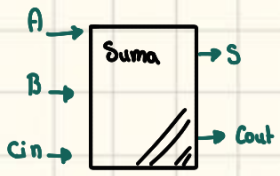
## Caja Negra

Identificamos entradas

- A
- B
- Cin

Identificamos Salidas

- S
- Cout



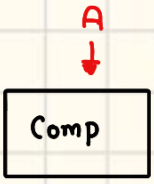

## Para el camino de datos

Cada accion se convierte en un

Registros A, B, Cin

Comparador Ai == Bi

Multiplexor S So

Contador i

Comparador 1 N